A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 11

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1 Subsistence maize:** 9.0 / 0.6 = 15 **1 Intensive wheat:** 54.0 / 18.0 = 3 **1 Feedlot beef:** 12.0 / 40.0 = 0.3 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. **Do not credit** • A bare wrong answer with no working.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** An energy subsidy is the energy put into a farming system by people rather than by the Sun: fuel for machinery, the manufacture of fertilisers and pesticides, irrigation, transport, drying and storage **1** Intensive systems raise the yield per hectare by adding large energy subsidies, so the output rises but the input rises faster and the ratio falls **1** The feedlot ratio is below 1, so it takes more energy to produce the beef than the beef contains – the extra trophic level means most of the energy in the feed is lost in respiration, excretion and the inedible parts of the animal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
01.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Keep the food chain short by producing crops for direct human consumption rather than feeding them to livestock, so one transfer of about 10% efficiency is avoided **1** Restrict the movement of livestock, so less of the energy in the feed is used for muscular activity and more is converted to biomass **1** Keep the animals warm, or rear them indoors, so less energy is spent maintaining body temperature **1** Slaughter the animals young, while their growth rate and food conversion ratio are still high **1** Use species or breeds with a high food conversion ratio, such as poultry or farmed fish, rather than cattle **1** Control parasites and disease, and use feed that is easy to digest, so more of the food eaten is assimilated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **food conversion ratio** → the feed needed per unit of food produced **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

Do not write outside the box

Question	Marking guidance	Marks
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** A single desired gene can be transferred, so the characteristic appears without the unwanted genes that come with a whole parent in conventional breeding **1** Genes can be moved between species that could never interbreed, so characteristics can be introduced that selective breeding could never produce, and the result is achieved in one generation instead of many **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
01.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Genes may escape by pollen transfer to wild relatives or to neighbouring non-GM crops, creating herbicide-resistant weeds or making organic certification impossible **1** Insect-resistant crops may kill non-target species, including the pollinators and natural pest predators that the wider ecosystem depends on **1** Pests and weeds evolve resistance to the toxin or the herbicide the crop is designed to be used with, so the advantage is temporary and use of the chemical increases **1** Reliance on a few patented GM varieties narrows the genetic diversity of the crop, making it more vulnerable to a new disease **1** Seed is patented and must be bought each year, which can put subsistence farmers in debt or make them dependent on one supplier **1** Long-term effects on human health are not fully known, and labelling may not let consumers choose **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **genetic diversity** → the range of alleles / genes present in a population **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Copying the structural / functional adaptations of living organisms in the design of manufactured items **1** Aircraft wingtip fins copied from splayed wingtip feathers of soaring birds, reducing turbulence and fuel use **1** Humpback whale flipper tubercles copied on ship rudders and wind turbine blades (tighter turns, rotation at lower wind speeds) **1** Shark skin ridges reducing friction / bacterial adhesion (surface coatings, hospital surfaces) **1** Termite mound convection copied for natural ventilation of buildings – hollow strutted bird bones in lightweight roofs – gecko toe adhesion – burr hooks - Velcro – lotus leaf self-cleaning glass – spider silk for airbags / body armour – bat / dolphin echolocation - ultrasound scanners **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	4
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Plants produce toxic chemicals (eg alkaloids) to defend themselves against herbivores – in controlled doses these affect human physiology **1 Examples:** morphine / codeine from poppies – quinine from cinchona bark (malaria) – aspirin from willow bark – taxol from yew (cancers) – diosgenin from Mexican yam (contraceptives, cortisone) **1 Animal examples accepted:** AZT from a marine sponge (HIV), shark liver lipids (tumours) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Only a very small proportion of species have been studied for medicinal substances so unknown species may contain valuable chemicals **1** Conserving the habitat protects all its species, including those not yet investigated (and their interdependent relationships) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.4	1 mark for the point below. **1** Squid giant nerve cells - nerve function / heart disease, stroke – marsupials (kangaroo, wallaby pouch) - embryo development – hippo skin secretion - sunscreen / burns – sea urchin embryos - teratogen testing – armadillo - leprosy – marine sponge proteins - transplant rejection **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-15; AQA spec 3.1.2.1.1-3.1.2.1.2.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** The readings are precise but may not be accurate **Also accept** • **precise** → readings close together / fine resolution • **accurate** → close to the true value	1
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** A result that differs from other results with which it was expected to be similar **1 Repeat the study / measurement:** if it recurs it is a 'real' result caused by another variable – if not it was an error and can be excluded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anomalous** → an outlier / a result that does not fit the pattern **Do not credit** • discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** To test the reliability of the method / equipment **1** To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B **Also accept** • **abiotic** → non-living / physical factors	1

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Deforestation **1** Ploughing of grassland / expansion of farmland **1** Reservoir creation / flooding **1** Mineral extraction, especially open-cast mining **1** Urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	
05.1	**1 mark each** 1 mark for each of the 4 points below. **1** Very small gene pool so individuals are closely related and inbreeding is likely **1** Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low **1** A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) **1** Exchange of individuals / semen between zoos – avoid hybridisation with related species / subspecies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4

Do not write outside the box

Question	Marking guidance	Marks
05.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided **1 Population interactions:** mixing breeding and non-breeding individuals causes conflict – some need a choice of partners, others pair for life **1 Breeding habitat needed:** courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) **1** Stress / disturbance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
05.3	**1 mark each** 1 mark for each of the 3 points below. **1 AI:** semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet – avoids dangerous mating (elephants, rhinos) and partner rejection **1 Embryo transfer:** a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) **1** So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** The new opportunity is overwhelmingly attractive (economic benefit) **1** Lack of knowledge / understanding of the impacts **1** The impacts occur elsewhere or much later, so are not evident to those who benefit **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Cultural belief that resources should be exploited to satisfy needs only **1** Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Resources are brought in from other areas **1** If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.4	**1 mark each** 1 mark for each of the 3 points below. **1** Easier access to energy has made many new activities possible **1** The scale of activities has greatly increased **1** Toxic wastes are produced that cannot be decontaminated by natural processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1.	
07.1	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Hedgerows **1** Hay meadows **1** Grazed moorland / heathland **1** Chalk grassland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive **1** Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2

Do not write outside the box

Question	Marking guidance	Marks
07.3	1 mark for the point below. **1** The introduced (non-indigenous) predator may eat other species as well as the pest **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
08.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** The size of the population, and the number of mature individuals able to breed **1** The rate of change of the population: how fast numbers have fallen, usually over the last ten years or three generations **1** The geographical range the species occupies and how far it is fragmented or dispersed, since a small or divided range makes extinction more likely **1** The quality and area of the remaining habitat, and whether the threats acting on the species are continuing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
08.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (12 000 - 2 100) / 12 000 x 100 **1** = 82.5% (accept 82 - 83%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
08.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** It is a keystone species, so losing it would change the whole community and many other species would be lost with it **1** It is endemic to the area, so if it is lost there it is lost everywhere – a species found in many countries can be conserved elsewhere **1** It is a flagship or charismatic species that attracts public interest and funding, which also protects everything else living in its habitat **1** It is a genetic resource with known or likely value: a wild relative of a crop, or a source of a medicine **1** Conservation is practical and likely to succeed: the threats can be removed, and suitable habitat remains for a reintroduction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **keystone species** → a species whose loss changes the whole community • **endemic** → found naturally only in that one place **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
08.4	**1 mark each** 1 mark for each of the 3 points below. **1** Appendix I lists species threatened with extinction, for which commercial international trade is banned and only exceptional permits such as scientific exchange are issued **1** Appendix II lists species that are not yet threatened with extinction but may become so unless trade is controlled, so trade is allowed under licence and monitored **1** Where the threat is demand in distant wealthy markets, cutting off the market removes the incentive to poach, and the agreement acts in every signatory country rather than only where the habitat is protected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **CITES** → the treaty that controls international trade in endangered species **Do not credit** • 'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix	3

Do not write outside the box

Question	Marking guidance	Marks
08.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Sites of Special Scientific Interest are notified for their wildlife or geology, and damaging operations on them need consent from the conservation agency **1** National Nature Reserves and Local Nature Reserves are owned or managed for conservation, with management plans and wardens **1** The Wildlife and Countryside Act makes it an offence to kill, injure or take listed species, or to damage their breeding sites and resting places **1** Special Areas of Conservation and Special Protection Areas, and Ramsar sites for wetlands, give international protection to habitats and to migratory birds **1** Green belts, National Parks and Areas of Outstanding Natural Beauty restrict development and so protect habitat indirectly **1** Agri-environmental schemes pay farmers to manage hedgerows, field margins and meadows for wildlife on land that is still farmed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-32, 38-39; AQA spec 3.1.2.3.1-3.1.2.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2** Carnivores fed pellets of sandeels, herring, anchovy – overfishing sandeels reduces puffins and terns – reduced by more plant protein / vegetable oil in pellets or farming herbivores (tilapia) **2** High density spreads lice (which spread to wild fish) and infections – pesticides (organophosphates, pyrethroids, hydrogen peroxide) kill wildlife – antibiotics breed resistant bacteria – reduced by lower densities, single-age tanks with flow from young to old, wrasse as biological control, warm-water washes, circulating flow **2** Escaped farmed fish breed with wild fish introducing disadvantageous genes (bright colour), or non-indigenous escapees colonise as predators / competitors – reduced by better cage design, sterile triploid fish (pressure-treated eggs), species that cannot breed locally (Japanese oysters in UK) **2** Faeces and uneaten food decompose causing deoxygenation under cages – reduced by monitored feeding, siting cages in strong currents, effluent treatment / reed beds for freshwater farms, IMTA with seaweed and shellfish absorbing wastes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	8
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Predators are killed under licence to protect stock, reducing wild populations of protected / valued species **1** Better cage designs, netting, fencing, bird scarers exclude predators without killing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 356-359; AQA spec 3.5.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Protected areas:** Korup NP Cameroon (1260 km^2, 600 tree species, endemic Ancistrocladus korupensis with anti-HIV / antimalarial alkaloids) – Tumucumaque NP Brazil (39 000 km^2, adjoining French Guiana reserve) – Alto Maues – Rio Bravo Belize funded by charities, companies and ecotourism • Debt for nature swaps – sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation – ITTO aims (largely unmet) – CITES listing of timber species (Honduras mahogany Appendix II) • Ecotourism income (but visitor damage in Costa Rica, Malaysia) – community projects and employment – carbon sequestration payments • **Effectiveness:** protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict – drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest – climate change cannot be controlled locally • **Judgement:** combination of large protected areas, economic alternatives for local people and reduced international demand needed – overall loss continues **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **endemic** → found naturally only in that one place • **sustainable** → meeting present needs without stopping future generations meeting theirs • **CITES** → the treaty that controls international trade in endangered species	9
	Do not credit • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Protected areas are the core of rainforest conservation. Korup National Park in Cameroon protects 1260 km^2 with about 600 tree species, including the endemic *Ancistrocladus korupensis* whose alkaloids have anti-HIV and antimalarial activity; Brazil's Tumucumaque National Park covers 39 000 km^2 and adjoins a reserve in French Guiana; the Rio Bravo area in Belize is funded by charities, companies, ecotourism and selective logging of high-value mahogany that does not damage the forest long term. Debt-for-nature swaps have cancelled Guatemalan and Brazilian debt in return for protection, CITES listing restricts trade in Honduras mahogany, the ITTO sets sustainable timber aims, ecotourism and community employment give local people an income from standing forest, and carbon sequestration payments are emerging.	
	Where they are enforced, protected areas work, but they face illegal logging, mining, poaching, road building and conflict, and ecotourism itself has damaged sites in Costa Rica and Malaysia. The deeper problem is that the drivers of deforestation - population growth and international demand for beef, soya, palm oil, timber and minerals - originate outside the forest, and climate change threatens even well-protected areas; the ITTO's aims are largely unmet and overall loss continues.	
	The most effective schemes are those that combine large protected areas with economic alternatives for local people, such as Rio Bravo, but they can only slow the loss unless demand in importing countries also falls.	
	Weaker answer - Level 2 (4 marks)	
	Rainforests are protected by national parks where logging is banned, for example in Brazil. Debt-for-nature swaps mean a country's debt is cancelled if it protects its forest. Ecotourism brings money from tourists so the forest is worth more standing than cut down. Sustainable logging only takes some trees. These efforts have saved some areas but deforestation is still happening because of illegal logging and farming, so they have only been partly effective.	
	What makes the difference	
	• Level 2 covers the main types of effort and a fair conclusion, but the question says 'describe' - Level 3 gives named examples with figures (Korup, Tumucumaque, Rio Bravo) as the mark scheme expects.	
	• Level 3 explains *why* effectiveness is limited (drivers originate outside the forest, ITTO aims unmet, climate change) rather than just noting continued loss.	
	• The judgement in Level 3 identifies what works best and what would be needed; Level 2 stops at 'partly effective'.	
10.2	1 mark for the point below. **1** Carbon sequestration reducing climate change – genetic resources / medicines – rainfall regulation – global biodiversity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Temperate broadleaf woodland:** seasonality, deep soils, high biodiversity, resources, climate and soil regulation – deforestation, fragmentation, management change – SSSIs, ancient woodland protection, coppicing, mixed planting • **Tropical rainforest:** stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils – fuelwood, timber, agriculture, mining, reservoirs, climate change – protected areas, debt-for-nature swaps, sustainable exploitation • **Tropical coral reefs:** cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity – fisheries, erosion protection, medicines, tourism – damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species – MPAs, no-take zones, control of activities • **Deep-water reefs:** cold, dark, slow growth – research, fisheries – trawling, oil / gas, acidification – SACs • **Oceanic islands:** isolation, endemism, no mammal predators	25

Do not write outside the box

Question	Marking guidance	Marks
	– exploitation, introduced species, habitat change, sea level – eradications, control of development and visitors • **Mangroves:** halophytes, anaerobic mud – erosion protection, fisheries nurseries, sediment trapping – clearance for aquaculture / urban / ports, pollution, climate change – protection, replanting • **Antarctica:** extreme cold, seasonality, marine nutrients, ice – water store, albedo, carbon, research – climate change, ozone, tourism, fishing, minerals – Antarctic Treaty, CCAMLR, tourism and waste rules • **Synthesis:** features -> importance -> threats -> strategy – evaluation of success – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how human activities influence biodiversity, using examples from a range of different contexts.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Direct exploitation:** food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil) – deliberate eradication of predators, competitors, vectors, pests • **Changes in abiotic factors:** water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage – inactivity (stopping plagioclimax management) • **Changes in biotic factors:** introduced competitors, predators, pathogens, hybridisation – loss of pollinators, seed dispersers, keystone species – food chain impacts (sea otters, puffins, jellyfish) • **Habitat destruction:** deforestation, farmland expansion, urbanisation, mining, reservoirs – fragmentation and edge effects • **Contexts:** temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica	25
	– agriculture, fishing, forestry • **Positive influences:** habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	

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